Simple Homemade Thermoelectric Generator, Thermistor and Pressure Sensor.

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A thermistor can be easily made using copper oxide. Two clean copper wires were mounted on a board as shown in the picture above. One or both of the wires were heated red hot in a propane torch while separated. After cooling the wires were adjusted so that they lightly touch. An ohm meter connected between the two wires would measure a high resistance (typically 40 to 100k). Holding a flame under the point where the two oxidized wires touch, can make the resistance fall to 1k or less. After the heat is removed the resistance will rise back to the original high value.

This simple homemade device can also be used to sense pressure. Squeezing the two wires gently will make the resistance reading on the ohm meter drop in proportion to the applied pressure.

In addition, this device can actually produce a voltage by heating the end of one of the oxidized wires. A volt-ohm meter set to the lowest dc current range will easily show the voltage generated. This can produce more voltage than any thermocouple that I have yet made. A meter set on the lowest dc current range also serves as a very low voltage meter.

A typical thermocouple like the ones used in gas water heaters when heated with a flame,

can produce typically 20 to 30 millivolts. These two pieces of oxidized copper wire under similar conditions, can easily produce more than 300 millivolts. The gas water heater thermocouple however, has the advantage of being able to produce a much greater amount of current.

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